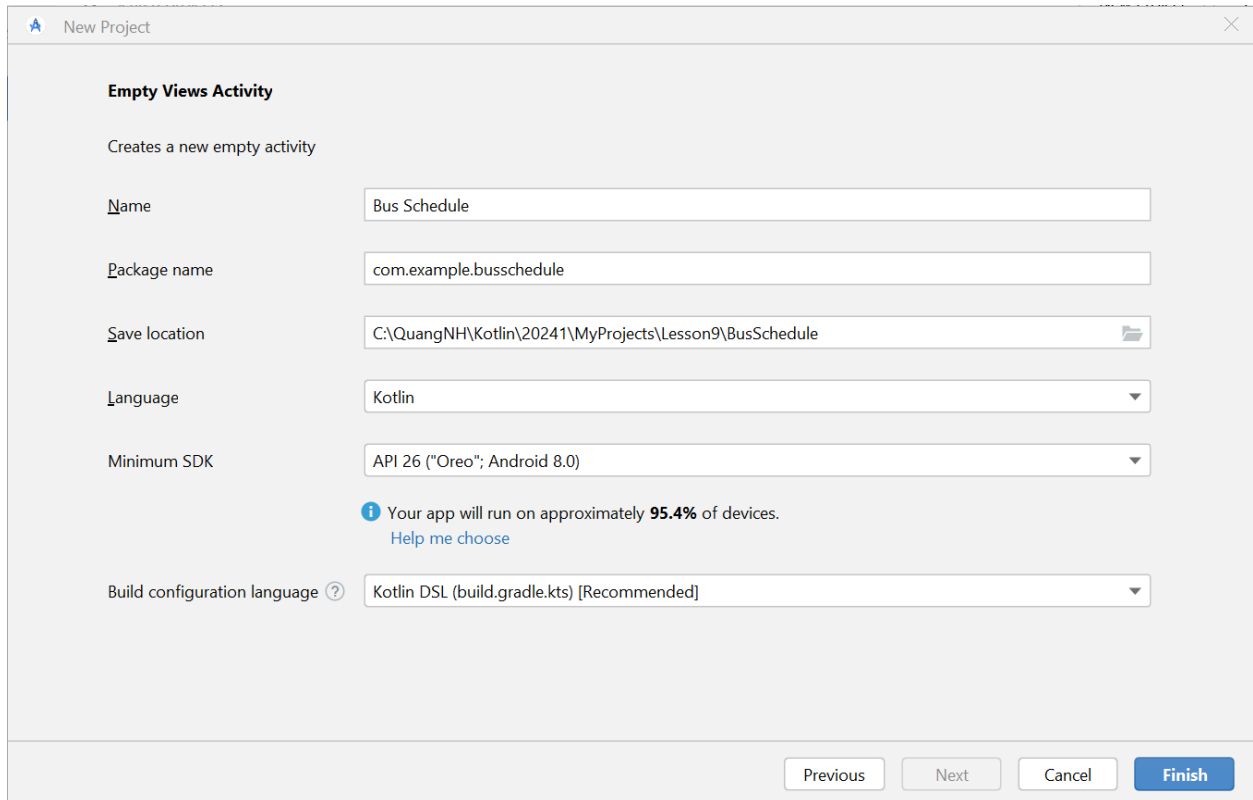


Lab 9.2. Make Starter code

Starter Code URL:

<https://github.com/google-developer-training/android-basics-kotlin-bus-schedule-app/tree/starter>

Bước 1. Tạo Project Bus Schedule



New Project

Empty Views Activity

Creates a new empty activity

Name: Bus Schedule

Package name: com.example.busschedule

Save location: C:\QuangNH\Kotlin\20241\MyProjects\Lesson9\BusSchedule

Language: Kotlin

Minimum SDK: API 26 ("Oreo"; Android 8.0)

i Your app will run on approximately **95.4%** of devices.
[Help me choose](#)

Build configuration language *?*: Kotlin DSL (build.gradle.kts) [Recommended]

Previous Next Cancel Finish

Bước 2. Cập nhật lên phiên bản SDK 35

Cập nhật trong file src/build.gradle.kts

```
android {  
    namespace = "com.example.busschedule"  
    compileSdk = 35  
  
    defaultConfig {  
        applicationId = "com.example.busschedule"  
        minSdk = 26  
        targetSdk = 35  
        versionCode = 1  
        versionName = "1.0"  
    }  
}
```

Bước 3. Cập nhật các thư viện cần thiết cho Navigation UI và Safe Args

Cập nhật trong file src/build.gradle.kts

```
plugins {  
    alias(libs.plugins.android.application) apply false  
    alias(libs.plugins.kotlin.android) apply false  
    id("androidx.navigation.safeargs") version "2.8.3" apply false  
}
```

Cập nhật trong file src/build.gradle.kts

```
plugins {  
    alias(libs.plugins.android.application)  
    alias(libs.plugins.kotlin.android)  
    id("androidx.navigation.safeargs.kotlin")  
}  
  
android {  
    namespace = "com.example.busschedule"  
    compileSdk = 35  
  
    defaultConfig {  
        applicationId = "com.example.busschedule"  
        minSdk = 26  
        targetSdk = 35  
        versionCode = 1  
        versionName = "1.0"  
  
        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"  
    }  
  
    buildTypes {  
        release {  
            isMinifyEnabled = false  
            proguardFiles(  
                getDefaultProguardFile("proguard-android-optimize.txt"),  
                "proguard-rules.pro"  
            )  
        }  
    }  
}
```

```

    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }
    kotlinOptions {
        jvmTarget = "1.8"
    }

    buildFeatures{
        viewBinding = true
    }
}

dependencies {
    implementation(libs.androidx.core.ktx)
    implementation(libs.androidx.appcompat)
    implementation(libs.material)
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)

    implementation("androidx.navigation:navigation-fragment-ktx:2.8.3")
    implementation("androidx.navigation:navigation-ui-ktx:2.8.3")
}

```

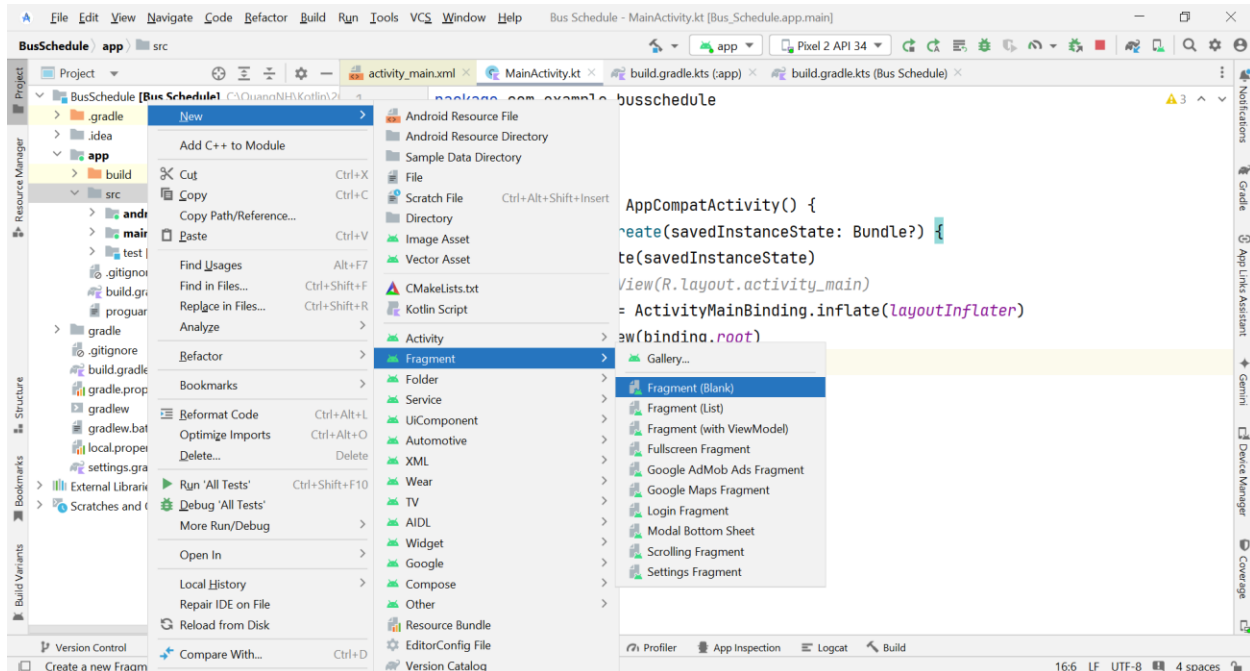
Bước 4. Cập nhật trong file MainActivity.kt

```

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        //setContentView(R.layout.activity_main)
        val binding = ActivityMainBinding.inflate(layoutInflater)
        setContentView(binding.root)
    }
}

```

Bước 5. Tạo FullScheduleFragment



Bước 6. Cập nhật file layout cho FullScheduleFragment

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<!--
```

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<http://www.apache.org/licenses/LICENSE-2.0>

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distributed under the License is distributed on an "AS IS" BASIS,

WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.

See the License for the specific language governing permissions and

limitations under the License.

-->

```
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".FullScheduleFragment"
    android:orientation="vertical">

    <TextView
        android:id="@+id/bus_stop_header"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="@string/bus_stop_header"
        android:textSize="16sp"
        android:gravity="center_horizontal"
        android:padding="8dp"
        app:layout_constraintWidth_percent="0.5"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="@id/arrival_time_header"
        app:layout_constraintEnd_toStartOf="parent"/>

    <TextView
        android:id="@+id/arrival_time_header"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
```

```

        android:text="@string/arrival_time_header"
        android:textSize="16sp"
        android:gravity="center_horizontal"
        android:padding="8dp"
        app:layout_constraintWidth_percent="0.5"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toEndOf="@id/bus_stop_header"
        app:layout_constraintEnd_toEndOf="parent"/>

```

```

<androidx.recyclerview.widget.RecyclerView
    android:id="@+id/recycler_view"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    app:layout_constraintTop_toBottomOf="@id/bus_stop_header"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

```

```

</androidx.constraintlayout.widget.ConstraintLayout>

```

Bổ sung các chuỗi ký tự vào /values/strings.xml

```

<resources>
    <string name="app_name">Bus Schedule</string>
    <!-- TODO: Remove or change this placeholder text -->
    <string name="hello_blank_fragment">Hello blank fragment</string>
    <!-- Shown above left column listing bus stops in full schedule fragment -->
    <string name="bus_stop_header">Stop Name</string>
    <!-- Shown above right column listing arrival times in full schedule fragment -->
    <string name="arrival_time_header">Arrival Time</string>
</resources>

```

Bước 7. Cập nhật file FullScheduleFragment.kt

```
package com.example.busschedule

import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView
import com.example.busschedule.databinding.FullScheduleFragmentBinding

class FullScheduleFragment : Fragment() {

    private var _binding: FullScheduleFragmentBinding? = null

    private val binding get() = _binding!!

    private lateinit var recyclerView: RecyclerView

    override fun onCreateView(
        inflater: LayoutInflater,
        container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View? {
        _binding = FullScheduleFragmentBinding.inflate(inflater, container, false)
        val view = binding.root
        return view
    }
}
```

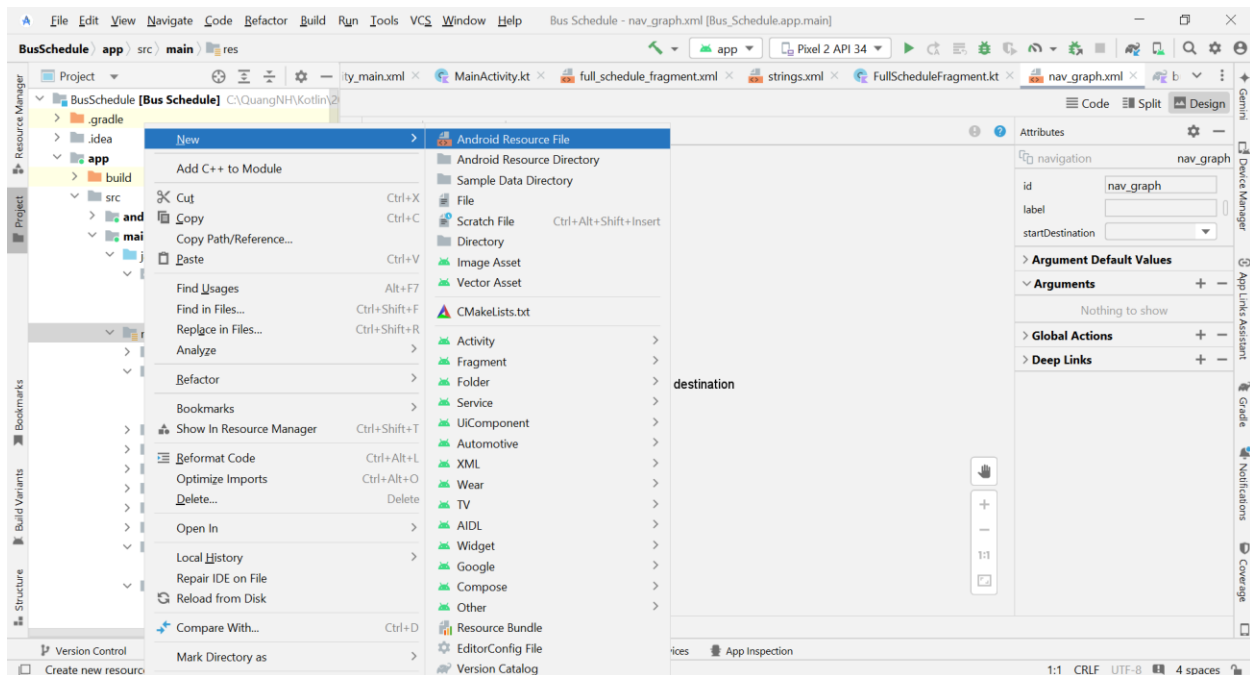
```
}
```

```
override fun onCreateView(view: View, savedInstanceState: Bundle?) {  
    super.onCreateView(view, savedInstanceState)  
    recyclerView = binding.recyclerView  
    recyclerView.layoutManager = LinearLayoutManager(requireContext())  
}
```

```
override fun onDestroyView() {  
    super.onDestroyView()  
    _binding = null  
}
```

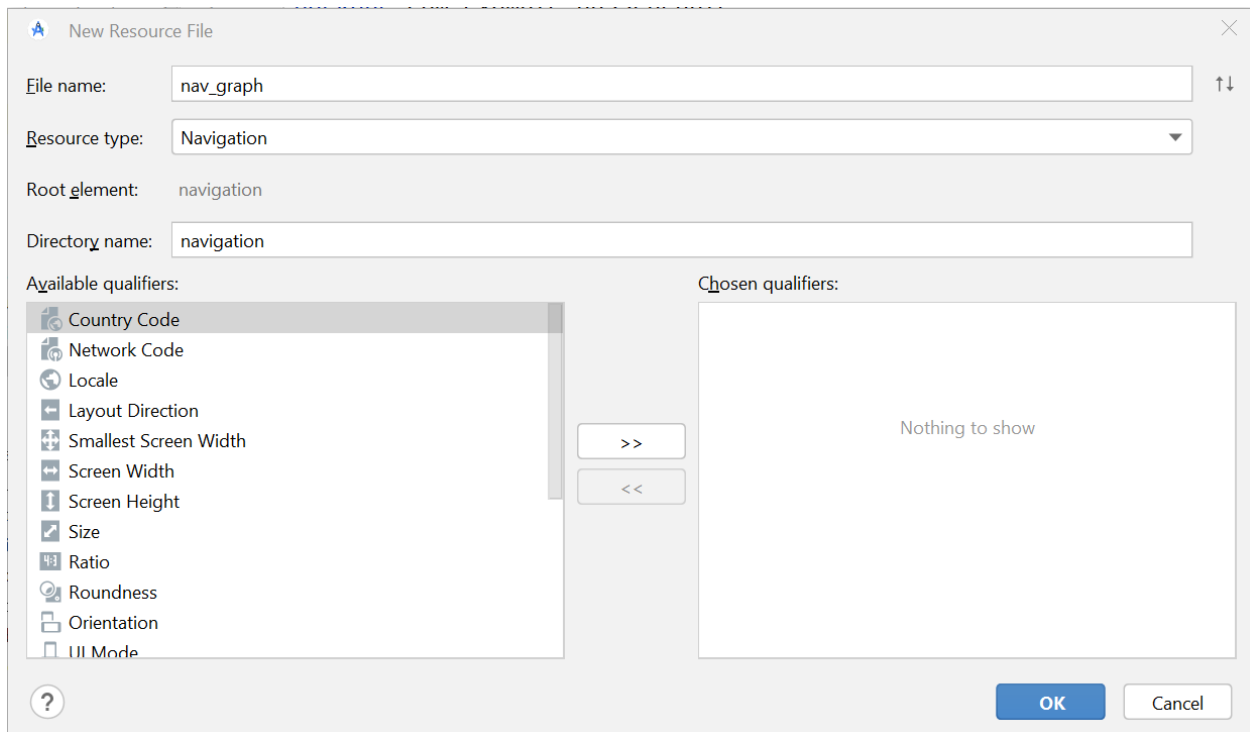
Bước 8. Bổ sung file navigation graph

Ấn chuột phải vào mục res => New => Android Resource File:

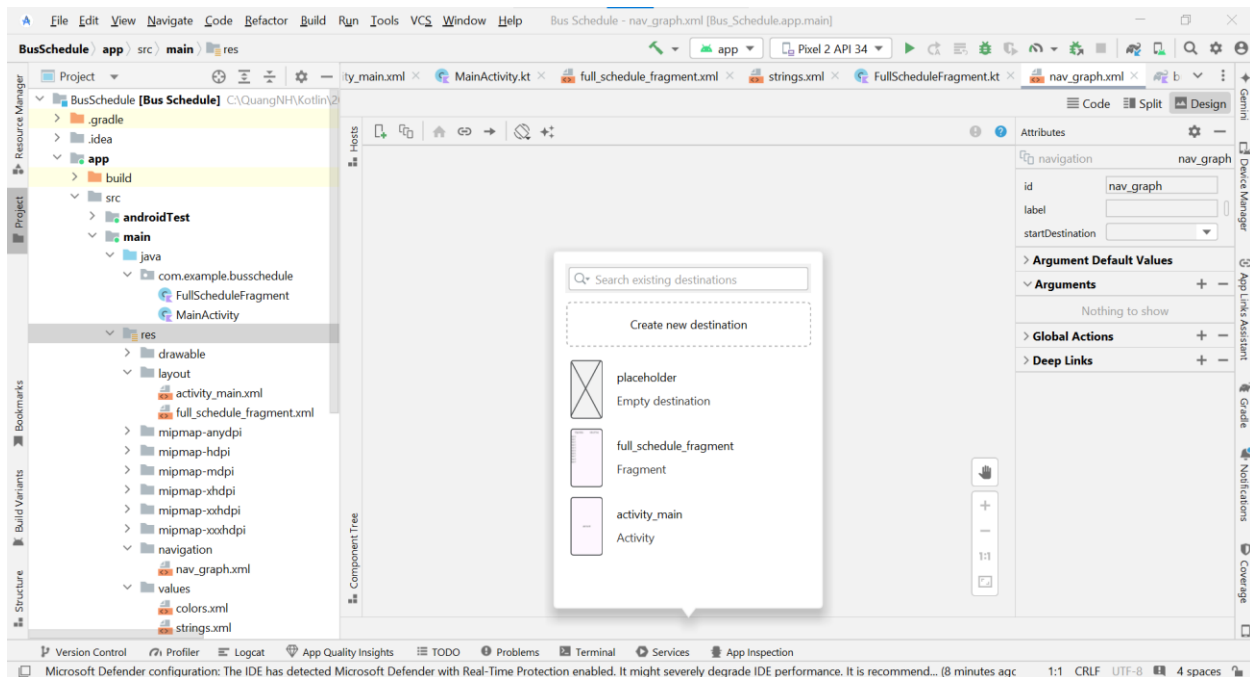


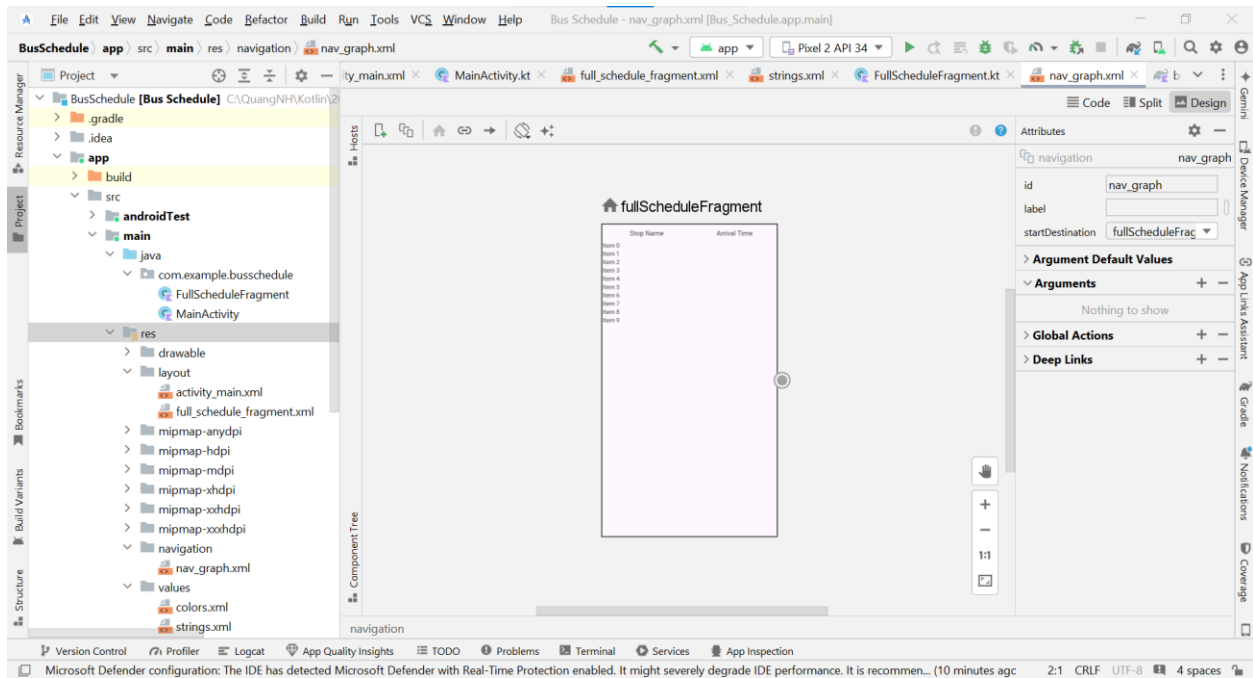
Trong cửa sổ “New Resource File”:

- Trong mục “Resource type”, chọn “Navigation”
- Trong mục “File name”, chọn: nav_graph



Bước 9. Thêm full_schedule_fragment vào navigation





Bước 10. Cập nhật file layout của MainActivity

Cập nhật file activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

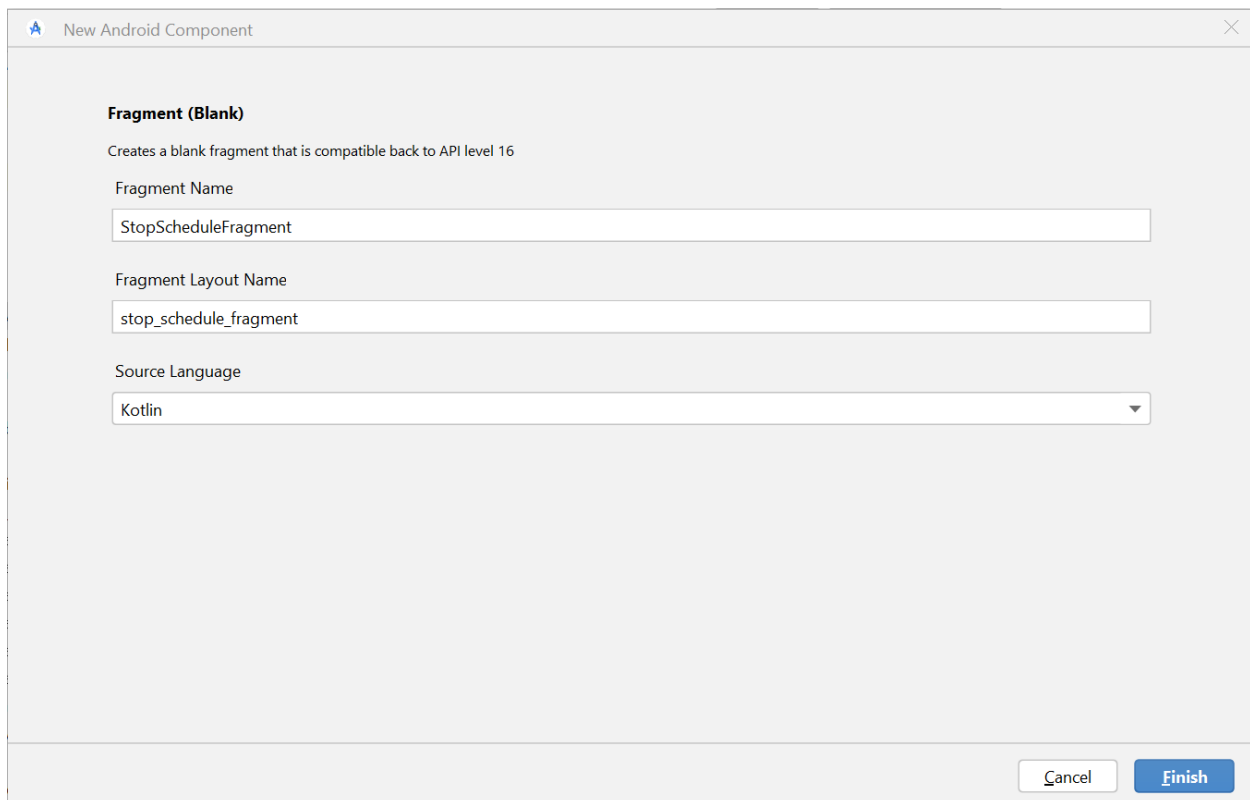
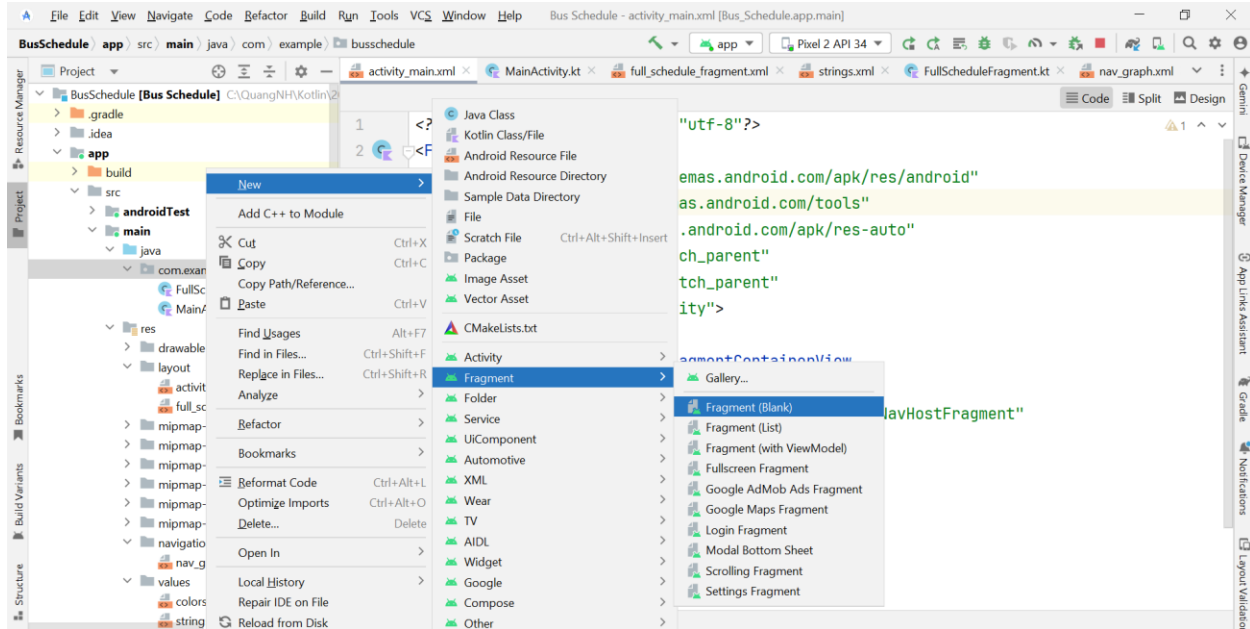
    <androidx.fragment.app.FragmentContainerView
        android:id="@+id/nav_host_fragment"
        android:name="androidx.navigation.fragment.NavHostFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:defaultNavHost="true"
        app:navGraph="@navigation/nav_graph"/>

</FrameLayout>
```

Chạy chương trình:



Bước 11. Thêm StopScheduleFragment



Bước 12. Cập nhật file StopScheduleFragment.kt

```
package com.example.busschedule
```

```
import android.os.Bundle
```

```
import androidx.fragment.app.Fragment
```

```
import android.view.LayoutInflater
```

```
import android.view.View
```

```
import android.view.ViewGroup
```

```
import androidx.recyclerview.widget.LinearLayoutManager
```

```
import androidx.recyclerview.widget.RecyclerView
```

```
import com.example.busschedule.databinding.StopScheduleFragmentBinding
```

```
class StopScheduleFragment : Fragment() {
```

```
    companion object {
```

```
        var STOP_NAME = "stopName"
```

```
    }
```

```
    private var _binding: StopScheduleFragmentBinding? = null
```

```
    private val binding get() = _binding!!
```

```
    private lateinit var recyclerView: RecyclerView
```

```
    private lateinit var stopName: String
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
arguments?.let {
    stopName = it.getString(STOP_NAME).toString()
}
}

override fun onCreateView(
    inflater: LayoutInflater, container: ViewGroup?,
    savedInstanceState: Bundle?
): View? {
    // Inflate the layout for this fragment
    _binding = StopScheduleFragmentBinding.inflate(inflater, container, false)
    val view = binding.root
    return view
}

override fun onViewCreated(view: View, savedInstanceState: Bundle?) {
    super.onViewCreated(view, savedInstanceState)
    recyclerView = binding.recyclerView
    recyclerView.layoutManager = LinearLayoutManager(requireContext())
}

override fun onDestroyView() {
    super.onDestroyView()
    _binding = null
}

}
```

Bước 13. Cập nhật file stop_schedule_fragment.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".StopScheduleFragment">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recycler_view"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</FrameLayout>
```

Bước 14. Thêm StopScheduleFragment vào navigation graph

```
<?xml version="1.0" encoding="utf-8"?>

<navigation xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/nav_graph"
    app:startDestination="@id/fullScheduleFragment">

    <fragment
        android:id="@+id/fullScheduleFragment"
        android:name="com.example.busschedule.FullScheduleFragment"
        android:label="full_schedule_fragment"
```

```

tools:layout="@layout/full_schedule_fragment" >
    <action
        android:id="@+id/action_fullScheduleFragment_to_stopScheduleFragment"
        app:destination="@id/stopScheduleFragment" />
    </fragment>
    <fragment
        android:id="@+id/stopScheduleFragment"
        android:name="com.example.busschedule.StopScheduleFragment"
        android:label="stop_schedule_fragment"
        tools:layout="@layout/stop_schedule_fragment" >
        <argument
            android:name="stopName"
            app:argType="string" />
        </fragment>
    </navigation>

```

Bước 15. Bổ sung Database vào project

Tạo thư mục assets/database trong thư mục main

Download file bus_schedule.db và copy vào thư mục này

https://github.com/google-developer-training/android-basics-kotlin-bus-schedule-app/blob/starter/app/src/main/assets/database/bus_schedule.db

Bước 16. Kết quả

